# Exercise list R9 — Mergers and acquisitions

Course topic 8 (ementário): the market for corporate control, takeovers and the free-rider problem (Grossman–Hart), toeholds and large shareholders (Shleifer–Vishny, Jegadeesh–Chowdhry), takeover defenses, means of payment under asymmetric information (Hansen, Fishman), LBOs and MBOs, acquisition valuation and the premium for control, and the equilibrium value of specialized assets (Shleifer–Vishny 1992). 15 real exercises ordered easy → really hard, copied verbatim from the course books for personal study; pages are printed, with the PDF page in brackets. None of them appears in another list. Links: exercise index · rules.

| Block | Exercises |
|---|---|
| Valuation of acquisitions, premium for control, EPS | 1, 6, 7 |
| Announcement returns and market efficiency | 3 |
| LBOs and MBOs | 2, 4 |
| Free riding, toeholds, large shareholders | 5, 10, 11 |
| Means of payment under asymmetric information | 8, 9, 12, 13 |
| Specialized assets, resale price, financial muscle | 14, 15 |

## References

| # | Tier | Book | Ch. | Exercise | Printed p. (PDF) | Answer in book | Trains |
|---|---|---|---|---|---|---|---|
| 1 | Easy | Higgins (Koski–Higgins–Mitton) 13e | Ch. 9 | 9.1 | 378 (399) | yes, pp. 425–426 | Five M&A facts: target returns, discount rate of the target, control premium, liquidation floor, MBO motive |
| 2 | Easy | Grinblatt & Titman 2e | Ch. 20 | 20.5 + 20.8 | 731 (744) | no | Which firms suit an LBO; why an MBO improves incentives without new managers |
| 3 | Easy | Higgins (Koski–Higgins–Mitton) 13e | Ch. 5 | 5.10 | 189 (210) | no | Target announcement returns, semistrong efficiency, price above the bid |

| # | Tier | Book | Ch. | Exercise | Printed p. (PDF) | Answer in book | Trains |
|---|---|---|---|---|---|---|---|
| 4 | Easy | Amaro de Matos | Ch. 7 | 7.2.4 #3 | 189 (207) | yes, p. 271 | Sources of the premiums paid to public shareholders in LBOs |
| 5 | Easy | Amaro de Matos | Ch. 8 | 8.1.5 #1 | 205 (223) | yes, p. 271 | Grossman–Hart free riding; a toehold makes a bid at $p \geq v$ profitable |
| 6 | Exam | Higgins (Koski–Higgins–Mitton) 13e | Ch. 9 | 9.3 | 379 (400) | yes, p. 426 | Cost of an acquisition (equity plus assumed debt) and the value placed on control |
| 7 | Exam | Higgins (Koski–Higgins–Mitton) 13e | Ch. 9 | 9.9 | 381 (402) | yes, p. 427 | "Avoid EPS dilution" as a merger criterion: maximum premium for low- and high-P/E targets |
| 8 | Exam | Amaro de Matos | Ch. 8 | 8.2.6 #3 | 223 (241) | yes, p. 276 | Hansen: adverse selection kills cash offers; a stock offer trades if $x < 150$ |
| 9 | Exam | Amaro de Matos | Ch. 8 | 8.2.6 #1 | 223 (241) | yes, p. 276 | Fishman (1988) preemptive bidding: why the signal is only partially revealing |

| # | Tier | Book | Ch. | Exercise | Printed p. (PDF) | Answer in book | Trains |
|---|---|---|---|---|---|---|---|
| 10 | Exam | Amaro de Matos | Ch. 8 | 8.1.5 #8 | 205 (223) | yes, pp. 274–275 | Jegadeesh–Chowdhry: the toehold as a signal breaks the pooling equilibrium |
| 11 | Exam | Tirole | Ch. 11 | 11.1 | 441 (454) | no | Poison pill (flip-over plan) when the new shares vote: control threshold, premium, optimal pill |
| 12 | Really hard | Amaro de Matos | Ch. 8 | 8.2.6 #4 | 223 (241) | no (p. 277 only cites page 80 of Hansen 1987) | Hansen: every feasible cash offer is dominated by a stock offer |
| 13 | Really hard | Amaro de Matos | Ch. 8 | 8.2.6 #8 | 223 (241) | yes, pp. 277–278 | Fishman (1989): cash signals a high valuation at a lower face value than debt |
| 14 | Really hard | Tirole | Ch. 14 | 14.2 | 515 (528) | yes, p. 618 | Shleifer–Vishny with an arbitrary resale price: redistribution, strategic complementarity, debt capacity |
| 15 | Really hard | Tirole | Ch. 14 | 14.3 | 515 (528) | yes, pp. 618–619 | Financial muscle with a random retooling cost: credit line, make-or-buy $P + \rho^* = 1$, monopoly price |

## Tier 1 — Easy

### 1. True or false: valuation and acquisitions

**Source:** Higgins (Koski–Higgins–Mitton), Ch. 9, Exercise 9.1, p. 378 (PDF 399) · Answer in book: yes, pp. 425–426

Is each of the following statements true or false? Explain your answers briefly.

a. On average, acquisitions destroy value for shareholders of acquired firms.

b. A discounted cash flow valuation of a target company discounts the target's estimated free cash flows at the acquirer's cost of capital.

c. An acquirer should be willing to pay a higher control premium for a well-managed company than a poorly managed one.

d. The liquidation value of a company's shares always places a floor under its stock price.

e. An unusually low stock price in managements' eyes encourages management to take the company private in a management buyout.

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### 2. LBO candidates and management buyouts

**Source:** Grinblatt & Titman, Ch. 20, Exercises 20.5 and 20.8, p. 731 (PDF 744) · Answer in book: no

**20.5.** Leveraged buyouts are observed mainly in industries with relatively stable cash flows and products that are not highly specialized. Explain why.

**20.8.** One of the stated benefits of a management buyout is the improvement in management incentives. In many cases, however, the top managers do not change after the buyout. Explain why.

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### 3. Target stock prices around acquisition announcements

**Source:** Higgins (Koski–Higgins–Mitton), Ch. 5, Exercise 5.10, p. 189 (PDF 210) · Answer in book: no

Suppose in Figure 5.3 that the stock prices of target firms in acquisitions responded to acquisition announcements over a three-day period rather than almost instantly.

a. Would you describe such an acquisition market as efficient? Why or why not?

b. Can you think of any trading strategy to take advantage of the delayed price response?

c. If you and many others pursued this trading strategy, what would happen to the price response to acquisition announcements?

d. Some argue that market inefficiencies contain the seeds of their own destruction. In what ways does your answer to this problem illustrate the logic of this statement, if at all?

e. Immediately after some merger announcements, the stock price of the target firm jumps to a level higher than the bid price. Is this proof of market inefficiency? What might explain this price pattern?

**Context needed to solve it** *From Higgins Ch. 5, "How Rapidly Do Stock Prices Adjust to New Information?", p. 168 (PDF 189):*

Figure 5.3 gives an indication of the speed with which common stock prices adjust to new information. It is a result of what is known as an event study. In this instance, the researchers are studying the effect of acquisition offers on the stock prices of target firms. The graph plots the cumulative percentage change in a target firm's stock price from a period beginning 40 days before the announcement of the acquisition offer and ending 10 days after. The price changes shown are averaged over 6,150 acquisition announcements in the U.S. between 1980 and 2008. The date for each announcement is defined as day 0, so that all announcements can be shown on the same timeline.

An acquisition offer is invariably good news to the target firm's shareholders, because the offer is at a price well above the prevailing market price of the firm's shares, so we expect to see the target company's stock price rise after

the announcement. The question is: How rapidly? The answer evident from the graph is: Very rapidly. We see that the stock price drifts upward prior to the announcement (perhaps due to rumors of a pending announcement), shoots up dramatically on the announcement day, and then drifts with little direction after the announcement. […]

*Figure 5.3, p. 169 (PDF 190), "Average Stock Price Reaction to Takeover Announcements for 6,150 Target Firms" (source: Betton, Eckbo, Thompson and Thorburn, Journal of Finance, 2014), transcribed from the PDF image. Horizontal axis: days relative to takeover announcement (Day 0), from −45 to 15; vertical axis: cumulative percent change in stock price of takeover target, 0 to 35. Values read from the curve (approximate):*

| Day | −40 | −30 | −20 | −10 | −5 | −1 | 0 | +1 | +5 | +10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Cumulative change (%) | 0 | ≈ 0.5 | ≈ 2 | ≈ 4 | ≈ 6 | ≈ 8–9 | ≈ 24 | ≈ 28.5 | ≈ 28 | ≈ 28 |

The curve rises slowly from day −40 to day −1, jumps almost vertically between day −1 and day +1, and is flat (slightly drifting down) from day +1 to day +10.

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**4. Sources of LBO premiums**

**Source:** Amaro de Matos, Ch. 7, Exercise 7.2.4 #3, p. 189 (PDF 207) · Answer in book: yes, p. 271

3. Empirical evidence is consistent in finding large positive premiums for public shareholders and also finding that these gains are substantially sustained. Explain some of the sources of such large gains.

**Context needed to solve it** *From Amaro de Matos §7.2.3 (empirical evidence on LBOs), pp. 187–189 (PDF 205–207) — the evidence the question summarizes:*

[…] Kaplan showed that the returns to the prebuyout shareholders are positively and significantly related to tax benefits from LBOs. The returns to the postbuyout shareholders, however, are not. They were shown to be positively related to operating income subsequent to the transaction. This result was seen by Kaplan as evidence that the tax benefits are mostly captured by the prebuyout shareholders. Besides, the hypothesis that returns to the shareholders are at the expense of bondholders is rejected. […]

The LBOs were described as some sort of concentration of ownership and control. Managers become owners of the firm, aligning their goals with those of maximizing the firm's value. However, Baker, Jensen, and Murphy (1988) show that actual executive compensation is very often insufficient to provide optimal incentives for managers. However, even if that is true, there is still room for reduction of agency costs in a different way, in particular following Jensen's free cash flow hypothesis. Increasing debt commits future cash flows, reducing managerial discretion in the allocation of free cash flow. […]

Regarding the control structure of LBOs, Kaplan (1989a) shows that the management postbuyout ownership increases by about three times the prebuyout values. […] More than two-thirds of the firms disclosed at least one restructuring activity undertaken since the LBO. […] Kaplan (1989a) and Smith (1990) also provide evidence of improved operational results following LBOs. […]

Finally, there is the issue of whether the gains of the prebuyout shareholders come at the expense of the prebuyout bondholders. Travlos and Cornett (1993) report significant losses at the statistical level for bondholders and preferred stockholders at the time of the LBO announcement. These losses are small, however, when compared to the gains to the prebuyout shareholder, thus failing to provide a satisfactory explanation.

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**5. Grossman–Hart with a toehold**

**Source:** Amaro de Matos, Ch. 8, Exercise 8.1.5 #1, p. 205 (PDF 223) · Answer in book: yes, p. 271

The next four exercises are related to Grossman and Hart (1990).

1. Show that if the bidder owns some shares of the targeted firm, it is still profitable to bid when $p \geq v$.

**Context needed to solve it** *From Amaro de Matos §8.1.1 "Largely Diffused Ownership", pp. 193–194 (PDF 211–212):*

A first model of takeovers is the one by Grossman and Hart (1990), who introduce the free-rider problem. Their idea is to formalize the way in which shareholders free-ride on the bidder potential improvements to the corporation, reducing the bidder's potential profits to zero and thus making takeovers unattractive.

Let $A$ denote the space of possible activities of the management, and $f(a)$ be the market value of the shares described as a function of some $a \in A$. The function $f$ describes to a certain extent the ability of the current management under which the maximum possible profit is

$$\max_{a \in A} f(a) = f(a^*).$$

Suppose now that a raider wishes to take over the firm and announces a tender price $p > f(a^*)$ at which he/she will buy all available shares. His/her abilities are supposed to be different from the current management and he/she is supposed to be able to improve the profits by an amount of $Z$. In this case, the expected market value is denoted by

$$v = \max_{a \in A} f(a) + Z,$$

where $Z > 0$. Assume that $v$ and $Z$ are known to all the shareholders. Clearly, if $v > p > f(a^*)$, the offer looks attractive to the shareholders and to the raider. However, in a diffusely held firm, such an attempt at control will not succeed. Here is an important assumption to reach this conclusion:

**Assumption 8.1.1.** *The ownership of the firm is largely diffused in the sense that there are a large number of shareholders, each of them holding a very small proportion of the outstanding shares.*

Therefore, given a tender offer, the impact of each shareholder's tender decision is negligible in the final result. Under rational expectations, it turns out that the outcome of the bid is not stochastic. That is to say, the only successful bids are those expected to be successful with a probability of 1 and all other bids will not happen. This is so, since given a tender offer with $p < v$, any shareholder who knows that the offer will succeed prefers to make $v$ instead of getting only $p$. Knowing that his/her decision has no impact on the outcome of the bid, each shareholder will be better off not tendering (thus receiving $p$) than tendering and receiving $v$. Therefore, no one tenders, and the bid does not succeed. The alternative would be to tender at $p \geq v$, but such conditions are clearly uninteresting to the bidder since his/her profit is reduced at least to zero.

Takeovers do exist, however. A way to explain them in this context is through different valuations due to information asymmetries, for instance. In such a case the shareholders would use, say, $v_s \neq v$ instead of $v$, and the bid could be successful at the tender price $v_s$ at which the shareholders would be indifferent between tendering and not tendering. If the cost of the raid is $c$, the raider's profit in this case would be

$$\Pi_r = v - v_s - c. \qquad (8.1)$$

*(Note: the parenthetical payoffs in "better off not tendering (thus receiving p) than tendering and receiving v" are swapped in the book; the argument requires that a non-tendering shareholder keeps a share worth v and a tendering one receives p.)*

## Tier 2 — Exam level

### 6. News Corp. and Dow Jones: cost of the acquisition and value of control

**Source:** Higgins (Koski–Higgins–Mitton), Ch. 9, Exercise 9.3, p. 379 (PDF 400) · Answer in book: yes, p. 426

In July 2007, News Corp. entered into an agreement to purchase all of the outstanding shares of Dow Jones and Company for $60 per share. Immediately prior to the News Corp. bid, the shares of Dow Jones traded at $33 per share. The number of outstanding shares at the time of the announcement was 82 million. The book value of interest-bearing liabilities on the balance sheet of Dow Jones was $1.46 billion.

a. Estimate the cost of this acquisition to the shareholders of News Corp.

b. What value did News Corp. place on the control of Dow Jones and Company?

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**7. Procureps: EPS dilution as a merger criterion**

**Source:** Higgins (Koski–Higgins–Mitton), Ch. 9, Exercise 9.9, p. 381 (PDF 402) · Answer in book: yes, p. 427

Procureps, Inc. (P) is considering two possible acquisitions, neither of which promises any enhancements or synergistic benefits. V1 is a poorly performing firm in a declining industry with a price-to-earnings ratio of 8 times. V2 is a high-growth technology company with a price-to-earnings ratio of 35 times. Procureps is interested in making any acquisition that increases its current earnings per share. All of Procureps's acquisitions are exchange-of-share mergers.

a. Calculate the maximum percentage premium Procureps can afford to pay for V1 and V2 by replacing the question marks in the following table.

b. What do your answers to part (a) suggest about the wisdom of using "avoid dilution in earnings per share" as a criterion in merger analysis?

| Company | P | V1 | P + V1 | V2 | P + V2 |
|---|---|---|---|---|---|
| Earnings after tax ($ millions) | 2 | 1 | 3 | 1 | 3 |
| Price-to-earnings ratio (X) | 30 | 8 | | 35 | |
| Market value of equity ($ millions) | ? | ? | | ? | |
| Number of equity shares (millions) | 1 | 1 | ? | 1 | ? |
| Earnings per share ($) | 2 | 1 | 2 | 1 | 2 |
| Price per share ($) | ? | ? | | ? | |
| Maximum new shares issued (millions) | | | ? | | ? |
| Value of new shares issued ($ millions) | | | ? | | ? |
| Maximum acquisition premium (%) | | | ? | | ? |

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**8. Hansen: cash versus stock with a uniform target value**

**Source:** Amaro de Matos, Ch. 8, Exercise 8.2.6 #3, p. 223 (PDF 241) · Answer in book: yes, p. 276

3. In the model of Hansen (1987), let $v$ be uniformly distributed over (0, 100) and take $w(v) = 1.5v$. Show that no mutually beneficial cash trade exists, but that a stock deal will work if $x < 150$.

**Context needed to solve it** *From Amaro de Matos §8.2.2 "Choosing the Means of Payment", pp. 207–209 (PDF 225–227):*

In a model developed by Hansen (1987), payment is made either in cash or in stock of the merged corporation. The basis of the mechanism that makes management and bidder prefer one or the other (either cash or stock) is asymmetry of information.

The acquirer owns a firm of value $x$, which is known to both the acquirer and the target's management. The target firm is worth $v$ to itself, but has a value $w(v) \geq v$ to the acquirer. The value of a merger will therefore be $x + v$. The asymmetry enters assuming that the bidder does not know the value of the firm $v$, but the current management does. Then, the target will never accept an offer below that value in a cash trade. This creates an adverse selection problem, against which the acquirer can protect himself/herself, defining his/her optimal strategy not on unconditional expected profit, but on expected profit conditional on the offer being accepted.

The acquirer takes $v$ as a random variable with support on the closed interval $[v_l, v_h]$. For a cash offer $C$, the expected profit of the raider is an expectation conditional on the acceptance of the offer—that is to say, conditional on $v_l < v < C$. If $f(v)$ denotes the density of probability of $v$,

$$E(\Pi_r \mid C) = \int_{v_l}^{C} [w(v) - C] f(v)\, dv.$$

Denoting by $F(v)$ the distribution of $v$ and by $E(w \mid C)$ the expected value of the target to the acquirer conditional on sale,

$$E(w \mid C) = \frac{1}{F(C)} \int_{v_l}^{C} w(v) f(v)\, dv,$$

the expected conditional wealth gain can be rewritten as

$$E(\Pi_r \mid C) = F(C)[E(w \mid C) - C].$$

The optimal cash offer $C$ must satisfy the first-order condition

$$w(C) - C - \frac{F(C)}{f(C)} = 0$$

together with the obvious restrictions of positive expected profit

$$E(\Pi_r \mid C) \geq 0$$

and $v_l \leq C \leq v_h$. If $E(\Pi_r \mid C) < 0$, which is to say when $C > E(w \mid C)$, or if the solution is at $v_l$, no cash trade will take place.

But if the bidder offers stock of the merged firm (say, a proportion $\beta$), trade may still occur. From the point of view of the acquirer, such a trade is in the interest of the target with certainty if the final value held by the owners of the target is greater than or equal to $v$ for whatever current value $v \in [v_l, v_h]$. It will be in the interest of the target with probability $F(v^*)$ if the same inequality holds for any $v \in [v_l, v^*]$. If the management of the target knows the function $w$, this means that with probability $F(v^*)$ trade is good for the target if

$$\beta[x + w(v)] \geq v \quad \forall v \in [v_l, v^*]$$

or

$$\beta \geq \frac{v}{x + w(v)} \quad \forall v \in [v_l, v^*], \qquad (8.11)$$

which is equivalent to

$$\beta \geq \max_v \frac{v}{x + w(v)}.$$

Making Hansen's additional assumption that

$$\frac{dw}{dv} < \frac{x + w(v)}{v}$$

the preceding condition becomes simply

$$\beta[x + w(v^*)] \geq v^*.$$

More easily, the acquirer will be interested only if the value held by him/her after the transaction is greater than the current value. In other words,

$$(1 - \beta)[x + E(w \mid v^*)] \geq x. \qquad (8.12)$$

These two conditions hold simultaneously if

$$\frac{E(w \mid v^*)}{\beta} - E(w \mid v^*) \geq x \geq \frac{v^*}{\beta} - w(v^*),$$

implying that

$$x \leq \frac{w(v^*) - v^*}{v^* - E(w \mid v^*)} E(w \mid v^*).$$

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**9. Fishman (1988): why the preemptive signal is not fully revealing**

**Source:** Amaro de Matos, Ch. 8, Exercise 8.2.6 #1, p. 223 (PDF 241) · Answer in book: yes, p. 276

1. Explain why the signaling equilibrium suggested by Fishman (1988) is not fully revealing. If $v_1$ could be privately observed by bidder 2, how would that change the signaling equilibrium?

**Context needed to solve it** *From Amaro de Matos §8.2.1 "Competition between Bidders", pp. 206–207 (PDF 224–225):*

The first model to be described here along these lines is the one by Fishman (1988) in which there are two bidders (say, bidder 1 and bidder 2), who make successive bids until only one remains. The remaining bid is accepted by the target, provided that its value is greater than or equal to the value $v$, which is assumed to be reasonable by the target. Otherwise, there is no takeover. Here, $v$ is taken as the current market value of the target, assuming that there is no takeover.

Let bidder 1 be the first to make an offer. The costs $c_i$ for $i = 1, 2$ of the bid are the costs to the bidders of private observation of their own valuations $v_i$, $i = 1, 2$. Furthermore, it is assumed that the valuations are random variables such that $E(\tilde{v}_i) < v$. This means that bidders never make a bid without first observing whether the realization of their valuation is above $v$. If $p$ is the initial offer made by bidder 1, it must be such that $v \leq p \leq v_1$ and at the same time makes competition unattractive to bidder 2. In other words, the expected payoff from competing to bidder 2 should be nonpositive.

The payoffs can be expressed simply in terms of a variable $d$ that takes the value 0 if bidder 2 does not compete and takes 1 otherwise. For any two random realizations of $v_1$ and $v_2$ the payoff for bidder 1 is

$$\Pi_1(v_1, v_2, p, d) = v_1 - (1 - d)p - d \min[\max(p, v_2), v_1] - c_1 \qquad (8.9)$$

and for bidder 2 is

$$\Pi_2(v_1, v_2, p, d) = d[(v_2 - \min(v_2, v_1) - c_2]. \qquad (8.10)$$

Bidder 2 does not have access to $v_1$ as bidder 1 does not know $v_2$. But both of them know the densities of probabilities $f_i(v_i)$, which are assumed to be strictly positive in a closed interval $[l, h]$ and zero elsewhere. It is clear that if $p$ is very large, it will discourage bidder 2 from making a higher bid. If it is not too costly, the ideal for bidder 1 is to find the threshold above which bidder 2 will not even incur the cost of being informed of his/her valuation $v_2$. Denoting this threshold by $r$, it must be such that the expected payoff of bidder 2 conditional on $v_1 \geq r$ is zero:

$$\frac{\int_r^h E_{v_2} \Pi_2(v_1, \tilde{v}_2, p, 1) f_1(v_1)\, dv_1}{\int_r^h f_1(v_1)\, dv_1} = 0.$$

For any bid $v'$ higher than $r$, the expected payoff of bidder 2 will be negative. On the other hand, bidder 1 will be able to preempt only if $v_1 > r$. In the event that $v_1 \leq r$, his/her best alternative is to offer $v$. Therefore, if he/she preempts, the gain he/she gets for doing so at an offer price $p'$ is

$$E_{v_2}\Pi_1(v_1, \tilde{v}_2, p', 0) - E_{v_2}\Pi_1(v_1, \tilde{v}_2, v, 1) = E_{v_2}\min[\max(v, \tilde{v}_2), v_1] - p'.$$

Let the highest price that bidder 1 is willing to pay be denoted by $\bar{p}(v_1)$. It has to be such that the preceding gain is still positive. Noticing that the gain is decreasing in $p'$, it follows that

$$\bar{p}(v_1) = E_{v_2}\min[\max(v, \tilde{v}_2), v_1].$$

It can be shown that it is an equilibrium to bidder 1 to offer $\bar{p}(r)$ if $v_1 \geq r$ and $v$ otherwise, whereas for bidder 2 it is an equilibrium not to bid in the first case and to bid in the second. It is a signaling equilibrium, although it is not fully revealing.

This model implies several things. It is clear that the expected payoff of bidder 1 increases with the cost $c_2$, since it is easier to preempt the entry of bidder 2 ($r$ decreases). If the first bid can be lower, the expected payoff of the target's shareholders will also be lower. [...]

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**10. Jegadeesh–Chowdhry: no pooling equilibrium**

**Source:** Amaro de Matos, Ch. 8, Exercise 8.1.5 #8, p. 205 (PDF 223) · Answer in book: yes, pp. 274–275

8. In the discussion of a pooling equilibrium in the context of Jegadeesh and Chowdhry (1994), assume that $\alpha_m = 0.3$ and that there are two types of bidders with $Z_L = 40$ and $Z_H = 80$. Assume further that 60% are of the high-$Z$ type and that $P(\pi) = \ln(\pi)$. Conditional on this information, show that there is no pooling equilibrium.

**Context needed to solve it** *From Amaro de Matos §8.1.2 "The Role of a Large Shareholder", p. 196 (PDF 214) — notation shared by §8.1.4:*

[...] Let $\alpha$ be the proportion of the shares owned by L. Since he/she is a minority, $0 < \alpha < \frac{1}{2}$. Also, let $Z$ denote the random increase in the discounted market value generated by the substitution of an inefficient management. [...] Under these conditions, the bidder will move to make an offer if he/she can purchase $\frac{1}{2} - \alpha$ of the shares. The expected market value under the existing management being $q$, the tender price must be higher than $q$, say, $p = q + \pi$ [...].

*From Amaro de Matos §8.1.4 "The Optimal Size of* $\alpha$ *before a Takeover", pp. 201–204 (PDF 219–222):*

Until now, the number of shares held by L has been considered as given. However, according to the preceding reasoning, the proportion $\alpha$ is likely to affect the probability that a takeover is successful. Hence, it is a strategic variable, the optimal value of which can be established before a takeover attempt. This problem was formalized by Jegadeesh and Chowdhry (1994), assuming that the initial shareholdings is an endogenous variable to be determined so as to maximize the value of the firm, an assumption that replaces Assumption 8.1.2.

It is assumed that the bidder buys a fraction $\alpha$ of the shares in the open market at a current market price of zero. The value of $\alpha$ will be taken as a signal of his/her motivation to take the control. In his/her attempt to acquire a controlling interest, the bidder can purchase up to $\alpha_{\max}$ shares in the open market, where $\alpha_{\max} < 0.5$, and the actual number of shares $\alpha \in [0, \alpha_{\max}]$ that the bidder acquires in the open market is determined endogenously.

The risk-neutral shareholder will not tender unless the bid is at least as large as the expected improvements of the profits, where both the bidder's toehold and the bid are considered as a signal; that is,

$$\pi \geq E(Z \mid \alpha, \pi).$$

This equation expresses the so-called *free-rider condition*. The value of the bid for the bidder is therefore

$$G(\pi, \alpha; Z) = P(\pi)[0.5Z - (0.5 - \alpha)\pi], \qquad (8.6)$$

and his/her problem is to maximize this value,

$$\max_{\pi,\alpha} G(\pi, \alpha; Z) \text{ subject to } \pi \geq E(Z \mid \alpha, \pi).$$

To obtain insightful results, two additional assumption are made:

**Assumption 8.1.4.** *The probability of a successful bid is increasing and concave in $\pi$.*

[…]

**Assumption 8.1.5.** *The constraint imposed by the free-rider condition is binding for all bidder types.*

Formally,

$$Z > \hat{\pi}(\alpha_{\max}, Z) > \hat{\pi}(\alpha, Z).$$

[…] Their first result states that, in the absence of the free-rider constraint, the value of bidding high is increasing in the bidder's type $Z$. […] The second result says that, in the absence of the free-rider constraint, the cost of choosing any level of initial $\alpha$ is identical for the bidder regardless of his/her value of $Z$. […]

The type of equilibrium obtained in this bid is discussed in detail by Jegadeesh and Chowdhry. There will be no pooling equilibrium—that is, an equilibrium in which $\alpha$ has no value as a signal and $\hat{\pi}$ is equal for all the bidders.

To see why such an equilibrium would not hold, suppose there are two bidders, with different values of $Z$—say, $Z_L$ and $Z_H$ with $Z_L < Z_H$. Assume that the unconditional expected value of $Z$ is $E(Z) = Z_M$, such that $Z_L < Z_M < Z_H$.

Since in a pooling equilibrium the minimum value of the bid has to be the expected value of $Z$ in order to avoid the free-rider problem and $\alpha$ plays no signaling role, every bidder will choose to bid $Z_M$ with $\alpha_m$ in hand.

This equilibrium does not hold because the bidder with $Z_L$ is bidding above his/her valuation. His/her strategy to escape from that situation is to bid $Z_L$ with a lower value of $\alpha$ (say, $\alpha'$) to be chosen such that the other bidder does not have an incentive to mimic him/her. That is to say, $\alpha'$ must satisfy

$$G(Z_L, \alpha'; Z_H) = G(Z_M, \alpha_m; Z_H).$$

It may be shown that in such a situation the low-valuation bidder is strictly better off, which is to say

$$G(Z_L, \alpha'; Z_L) > G(Z_M, \alpha_m; Z_L)$$

and hence the pooling equilibrium does not survive. […]

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**11. Takeover defenses: a flip-over pill whose new shares vote**

**Source:** Tirole, Ch. 11, Exercise 11.1, p. 441 (PDF 454) · Answer in book: no

**Exercise 11.1 (takeover defenses).** Extend the analysis of takeover defenses in Section 11.5.2 to the case in which the new shares created by the flip-over plan carry a voting right.

**Context needed to solve it** *From Tirole §11.5 "Positive Theory of Takeovers: Single-Bidder Case", pp. 430–431 (PDF 443–444):*

[…] In a first step we will assume that all shares carry equal voting rights and that the raider needs a simple majority or, more generally, a fraction $\kappa \in (0, 1)$ of the shares in order to gain control, replace the incumbent management, and implement the new policy. That is, a raider who purchases only a minority of shares or, more generally, a fraction less than $\kappa$ of the shares is on the same footing as any other investor, and neither delivers investor value $\hat{v}$ nor enjoys rent $\hat{w}$ from control. […]

We say that the raider enhances value (to investors) if $\hat{v} > v$. The case $\hat{v} < v$ corresponds to a "value-decreasing raider." We focus on the case in which the value enhancement or decrease is symmetric information.

*($v$ is the investor value of the firm under incumbent management, $\hat{v}$ under raider management; §11.5.3, p. 434: "Each share carries a cash-flow right equal to $1/n$th of the investor payoff ($v$ under incumbent management, $\hat{v}$ under raider management).")*

*From Tirole §11.5.1 "Value-Enhancing Raider: The Grossman–Hart Analysis", pp. 431–432 (PDF 444–445):*

[…] let us normalize to 1 the value added by the raider's taking over:

$$\hat{v} - v \equiv 1.$$

And let us redefine $P$ as the premium over $v$ offered by the raider. That is, the raider offers price $v + P$. The relevant range for $P$ is $[0, 1]$. A negative-premium offer is always rejected, while premia above 1 are accepted but wasteful for the raider.

We will assume that there is a continuum, of mass 1, of shareholders. The assumption of a continuum finesses the issue of a shareholder's potentially being "pivotal," i.e., affecting the outcome through his tendering choice. A shareholder will therefore compare the takeover premium that is offered by the raider to the expected value enhancement, taking the probability of takeover as exogenous to his tendering choice and equal to its equilibrium value. […]

Consider an unrestricted, unconditional offer, and assume that the raider needs to acquire a fraction $\kappa \in (0, 1)$ of the shares to gain control. We claim that the probability of takeover success must be equal to the premium:

$$\beta \equiv \Pr(\text{takeover success}) = P.$$

If this probability exceeded $P$, then each shareholder would be better off holding on to his share, since $\beta\hat{v} + (1-\beta)v > v + P$; and so the takeover would fail with probability 1, a contradiction. Similarly, for a probability of success smaller than $P$, all would be better off tendering and thus the takeover would succeed. So the equilibrium probability of takeover success must equal the premium. The fraction of shares tendered must be exactly equal to $\kappa$. […] Leaving aside any private surplus $\hat{w}$, the profit made by the raider on the takeover attempt is

$$\pi = \kappa[\beta \cdot 1 - P] = 0.$$

That is, the raider is unable to derive any benefit from the value enhancement. […]

*From Tirole §11.5.2 "Positive Raider Surplus despite Free Riding", pp. 432–434 (PDF 445–447):*

**11.5.2.1 Private Benefit from Control.** When the raider derives a private surplus $\hat{w}$ from control, then he gets to keep this surplus and optimally bids $P = 1$. To see this, note that the raider's profit when offering premium $P$ is

$$\pi = \kappa[\beta - P] + \beta\hat{w} = P\hat{w}.$$

The raider strictly prefers to bid the maximum premium: $P = 1$. […]

**11.5.2.2 Toehold.** […] Suppose that the raider already owns a fraction $\theta < \kappa$ of the shares when making a tender offer. Assuming, again, that $\hat{w} = 0$, the raider's profit for premium $P$ is

$$\pi(P) = (\kappa - \theta)(\beta - P) + \theta\beta,$$

where $\beta$ is, as earlier, the probability of takeover success and must in equilibrium be equal to $P$. Hence, $\pi(P) = \theta P$. The optimal bid is then $P = 1$, yielding profit $\pi = \theta$. […]

**11.5.2.4 Takeover Defenses.** Takeover defenses come in many guises, and, except for the common feature that they make it harder for a raider to acquire a firm, are hard to summarize concisely. Let us illustrate their role in the case of poison pills, more specifically in the most common form of a "flip-over plan" under which the holders of shares are entitled to purchase new shares at a substantial discount after a hostile takeover.

For computational simplicity, let us assume a simple majority rule ($\kappa = \frac{1}{2}$) and that the new shares carry no voting rights.[26] In the case of takeover success, the 50% of shares kept by the initial shareholders are worth $\hat{v} + \Delta$ (with

$\Delta > 0$), while the 50% acquired by the raider are worth $\hat{v} - \Delta$ to him due to the dilution.[27] Letting, as before, $\beta$ denote the probability of success and $P$ the premium over $v$, shareholders are indifferent between tendering and keeping their shares if and only if

$$\beta(\hat{v} + \Delta) + (1 - \beta)v = v + P \quad \text{or} \quad \beta = \frac{P}{1 + \Delta}.$$

The raider's profit is then

$$\pi = \beta\hat{w} + \tfrac{1}{2}[\beta(\hat{v} - \Delta) + (1 - \beta)v - (v + P)] = \beta(\hat{w} - \Delta).$$

Assuming that $\hat{w} > \Delta$ (otherwise the raider makes no offer), it is optimal for the raider to succeed for certain ($\beta = 1$) by choosing $P = 1 + \Delta$.

The poison pill further raises the purchase price. In contrast with the dilution of initial shareholders by the raider considered in Grossman and Hart, poison pills allow a dilution of the raider by initial shareholders. Poison pills thereby allow the firm to adjust the purchase price paid by the raider. Suppose, for example, that the raider's benefit from control, $\hat{w}$, is known (the distribution $H$ is a spike at $\hat{w}$).[28] In the absence of a poison pill, $P = 1$ and the raider's surplus is $\hat{w}$. The optimal poison pill then yields dilution $\Delta^* = \hat{w}$.

[26] If the new shares carry a voting right, the analysis is basically unchanged.

[27] For example, suppose that the holders of untendered shares are entitled to one extra share per share for free. Then the raider has only one-third of the cash rights. Then $\Delta = \frac{1}{3}\hat{v}$.

[28] And $v + w < \hat{v} + \hat{w}$.

## Tier 3 — Really hard

**12. Hansen: every cash offer is dominated by a stock offer**

**Source:** Amaro de Matos, Ch. 8, Exercise 8.2.6 #4, p. 223 (PDF 241) · Answer in book: no (p. 277 only cites page 80 of Hansen 1987)

4. Given asymmetric information in the model of Hansen (1987), show that for every feasible cash offer there exists a stock offer that is preferred by the acquirer.

**Context needed to solve it** *From Amaro de Matos §8.2.2 "Choosing the Means of Payment", pp. 207–210 (PDF 225–228):*

In a model developed by Hansen (1987), payment is made either in cash or in stock of the merged corporation. The basis of the mechanism that makes management and bidder prefer one or the other (either cash or stock) is asymmetry of information.

The acquirer owns a firm of value $x$, which is known to both the acquirer and the target's management. The target firm is worth $v$ to itself, but has a value $w(v) \geq v$ to the acquirer. The value of a merger will therefore be $x + v$. The asymmetry enters assuming that the bidder does not know the value of the firm $v$, but the current management does. Then, the target will never accept an offer below that value in a cash trade. This creates an adverse selection problem, against which the acquirer can protect himself/herself, defining his/her optimal strategy not on unconditional expected profit, but on expected profit conditional on the offer being accepted.

The acquirer takes $v$ as a random variable with support on the closed interval $[v_l, v_h]$. For a cash offer $C$, the expected profit of the raider is an expectation conditional on the acceptance of the offer—that is to say, conditional on $v_l < v < C$. If $f(v)$ denotes the density of probability of $v$,

$$E(\Pi_r \mid C) = \int_{v_l}^{C} [w(v) - C] f(v)\, dv.$$

Denoting by $F(v)$ the distribution of $v$ and by $E(w \mid C)$ the expected value of the target to the acquirer conditional on sale,

$$E(w \mid C) = \frac{1}{F(C)} \int_{v_l}^{C} w(v) f(v)\, dv,$$

the expected conditional wealth gain can be rewritten as

$$E(\Pi_r \mid C) = F(C)[E(w \mid C) - C].$$

[...] But if the bidder offers stock of the merged firm (say, a proportion $\beta$), trade may still occur. From the point of view of the acquirer, such a trade is in the interest of the target with certainty if the final value held by the owners of the target is greater than or equal to $v$ for whatever current value $v \in [v_l, v_h]$. It will be in the interest of the target with probability $F(v^*)$ if the same inequality holds for any $v \in [v_l, v^*]$. If the management of the target knows the function $w$, this means that with probability $F(v^*)$ trade is good for the target if

$$\beta[x + w(v)] \geq v \quad \forall v \in [v_l, v^*]$$

or

$$\beta \geq \frac{v}{x + w(v)} \quad \forall v \in [v_l, v^*], \qquad (8.11)$$

which is equivalent to

$$\beta \geq \max_v \frac{v}{x + w(v)}.$$

Making Hansen's additional assumption that

$$\frac{dw}{dv} < \frac{x + w(v)}{v}$$

the preceding condition becomes simply

$$\beta[x + w(v^*)] \geq v^*.$$

More easily, the acquirer will be interested only if the value held by him/her after the transaction is greater than the current value. In other words,

$$(1 - \beta)[x + E(w \mid v^*)] \geq x. \qquad (8.12)$$

[...] In a stock offer the expected wealth of the acquirer will be

$$F(v^*)(1 - \beta)[E(w \mid v^*) + x] + [1 - F(v^*)]x. \qquad (8.13)$$

[...] The conditions under which cash or stock trades with a given probability are possible are, therefore, established. Of course, it is possible that they are simultaneous. In such a case, for each feasible cash offer there will be a stock offer with the same probability of success that will be preferred by the acquirer. To see this, notice that the expected wealth of the acquirer after a cash offer $C$ equals

$$F(C)[E(w \mid C) - C] + x. \qquad (8.14)$$

Constructing a stock offer with $\beta$ given by

$$\beta^* = \frac{C}{x + w(C)},$$

it is known from (8.11) that it will have the same probability of success $F(C)$ as a cash offer of value $C$. The acquirer's expected wealth in this case will be

$$F(C)(1 - \beta^*)[E(w \mid C) + x] + [1 - F(C)]x.$$

It is a simple algebraic exercise to check that this expression is greater than the one in (8.14). Therefore, in the situation described up to now in this model, no cash offers will occur, and only stock trade is possible.

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**13. Fishman (1989): the face value needed to signal, cash versus debt**

**Source:** Amaro de Matos, Ch. 8, Exercise 8.2.6 #8, p. 223 (PDF 241) · Answer in book: yes, pp. 277–278

8. Show that, in the model of Fishman (1989), the face value needed to signal that $v_1$ is at or above any given level is lower with cash compared to debt.

**Context needed to solve it** *Fishman (1988) base model — from Amaro de Matos §8.2.1, pp. 206–207 (PDF 224–225):*

[…] there are two bidders (say, bidder 1 and bidder 2), who make successive bids until only one remains. The remaining bid is accepted by the target, provided that its value is greater than or equal to the value $v$, which is assumed to be reasonable by the target. Otherwise, there is no takeover. Here, $v$ is taken as the current market value of the target, assuming that there is no takeover.

Let bidder 1 be the first to make an offer. The costs $c_i$ for $i = 1, 2$ of the bid are the costs to the bidders of private observation of their own valuations $v_i$, $i = 1, 2$. Furthermore, it is assumed that the valuations are random variables such that $E(\tilde{v}_i) < v$. […] If $p$ is the initial offer made by bidder 1, it must be such that $v \leq p \leq v_1$ and at the same time makes competition unattractive to bidder 2. In other words, the expected payoff from competing to bidder 2 should be nonpositive.

The payoffs can be expressed simply in terms of a variable $d$ that takes the value 0 if bidder 2 does not compete and takes 1 otherwise. For any two random realizations of $v_1$ and $v_2$ the payoff for bidder 1 is

$$\Pi_1(v_1, v_2, p, d) = v_1 - (1 - d)p - d \min[\max(p, v_2), v_1] - c_1 \qquad (8.9)$$

and for bidder 2 is

$$\Pi_2(v_1, v_2, p, d) = d[(v_2 - \min(v_2, v_1) - c_2]. \qquad (8.10)$$

[…] both of them know the densities of probabilities $f_i(v_i)$, which are assumed to be strictly positive in a closed interval $[l, h]$ and zero elsewhere. […] Denoting this threshold by $r$, it must be such that the expected payoff of bidder 2 conditional on $v_1 \geq r$ is zero […].

*From Amaro de Matos §8.2.3 "Cash as a Preemptive Instrument with Many Bidders", pp. 212–214 (PDF 230–232):*

In the paper of Hansen, there is only one acquirer for the target. Fishman (1989) developed a model for choice of means of payment that allows more potential acquirers. In this model it is possible to see how the presence of competing bidders may affect the choice of the means of payment differently.

His idea was to extend his model of preempting bids discussed before, making the choice of the means of payment a reinforcing preemptive device.

The model is basically Fishman's earlier model, but the profit of the bidders will depend on an additional variable $\theta \in \{1, 0\}$ describing the chosen means of payment (1 for cash and 0 for issuing debt). In the event of a debt offer, $p$ will denote its face value.

Since the value of a debt offer is state contingent, it is assumed that the target costlessly observes a private signal $s$ that conveys information on both bidders' valuations. It is supposed that $s$ may assume only two possible values, $s^+$ and $s^-$, referring respectively to a "good" and a "bad" state of the world, with probabilities $\gamma$ and $1-\gamma$. The valuations of the bidders will also depend on the state of the world that they infer. In other words, it is as if the density functions $f_i$ depended on $s$. The value of any variable of interest will be labeled with a "+" if it is conditional on $s = s^+$ and with a "−" if conditional on $s = s^-$. For instance, it is assumed that $v_i^+ > v_i^-$. That is to say, for any bidder the valuation is higher in the good state of the world. Hence, the assumption that any bidder 1's offer face value is between $v$ and $v_1$ may read

$$v \leq p \leq v_1^+.$$

In particular, a debt offer with face value $p$ is worth $\min\{v_i^+, p\} \geq v$ if $s = s^+$ and $\min\{v_i^-, p\} < v$ if $s = s^-$. Thus, the target will accept the offer if $s = s^+$ and will reject it if $s = s^-$.

The main difference between a debt and a cash offer is that the value of the cash offer does not depend on the state of the world. Thus, cash offers cannot induce the target to make an efficient acceptance/rejection decision.

The profit itself will depend, directly and indirectly, on the state of the world. Introducing a variable $\delta$ that equals 1 when $s = s^+$ and 0 when $s = s^-$, the profits in (8.9) and (8.10) are replaced by

$$\Pi_1(v_1, v_2, \delta, p, \theta, d) = [\theta + (1-\theta)\delta]v_1 - [\theta(1-\delta) + \delta(1-d)]p - d\delta \min[\max(p, v_2), v_1] - c_1$$

$$\Pi_2(v_1, v_2, \delta, p, \theta, d) = d\{\delta[v_2 - \min(v_2, v_1)] - c_2\} \qquad (8.15)$$

for bidder 1 and bidder 2, respectively.

As before, bidder 1 tries to find the threshold $r$ for the firm value above which bidder 2 will have zero profit. In other words, $r$ is implicitly defined through

$$\frac{\int_r^h E_{v_2}\Pi_2(v_1, \tilde{v}_2, \delta, p, \theta, 1) f_1(v_1)\, dv_1}{\int_r^h f_1(v_1)\, dv_1} = 0.$$

For any bid $v'$ higher than $r$, the expected payoff of bidder 2 will be negative. On the other hand, bidder 1 will be able to preempt only if $v_1 > r$. Therefore, the expected gain of preempting at an offer price $p'$, other things constant, is

$$E\Pi_1(v_1, \tilde{v}_2, \delta, p', \theta, 0) - E\Pi_1(v_1, \tilde{v}_2, \delta, p', \theta, 1) = \gamma\{E_{v_2} \min[\max(p', \tilde{v}_2^+), v_1^+] - p'\}.$$

This expression is clearly increasing in $v_1$. Hence, high-valuing first bidders stand to lose more in an auction and thus have a greater incentive to deter competition.

On the other hand, the expected gain of making a debt offer compared to a cash offer at face value $p'$, other things constant, is

$$E\Pi_1(v_1, \tilde{v}_2, \delta, p', 0, d) - E\Pi_1(v_1, \tilde{v}_2, \delta, p', 1, d) = (1-\gamma)[p' - v_1^-].$$

This gain is decreasing in $v_1$. From these two results it appears that first bidders will tend to preempt with cash whenever they have a high valuation for the target, and will possibly make a debt offer if they do not intend to preempt. [...]

Now, let us cross these two pieces of information. Let $p_d(v_1)$ denote the highest face value of an offer that bidder 1 is willing to make in order to preempt. It has to be such that the preceding gain of preempting is zero. As we notice that the gain is decreasing in $p'$, it follows that

$$p_d(v_1) = E \min[\max(v, \tilde{v}_2^+), v_1^+].$$

As for a cash offer, its maximum value $p_c(v_1)$ is defined by a zero expected profit from both preempting and paying in cash, which is to say, $p_c(v_1)$ must satisfy

$$\gamma\{E_{v_2}\min[\max(p_c(v_1),\tilde{v}_2^+),v_1^+]-p_c(v_1)\}=(1-\gamma)[p_c(v_1)-v_1^-]$$

or

$$p_c(v_1)=\gamma E_{v_2}\min[\max(v,\tilde{v}_2^+),v_1^+]+(1-\gamma)v_1^-.$$

It is possible to show that there exists a perfect sequential equilibrium such that the optimal strategy of bidder 1 is to make a debt offer of value $v$ if he/she cannot preempt (which is to say, if $v\le v_1\le r$) or to make a cash offer with value $p_c(r)$ if preemption is possible (in other words, if $v_1>r$).

The optimal strategy of bidder 2, on the other hand, will be to compete if bidder 1 made a debt offer with face value lower than $p_d(r)$ or if he/she made a cash offer of value lower than $p_c(r)$. If bidder 1's cash offer equals (or is greater than) $p_c(r)$, bidder 2 does not compete.

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**14. Shleifer–Vishny with an arbitrary distribution of bargaining power**

**Source:** Tirole, Ch. 14, Exercise 14.2, p. 515 (PDF 528) · Answer in book: yes, p. 618

**Exercise 14.2 (alternative distributions of bargaining power in the Shleifer–Vishny model).** Perform the analysis of Section 14.2.2 for an arbitrary unit price $P\in[0,\rho_0]$ of resale of a distressed firm's assets to a productive one. (Assume that bargaining occurs between the two firms' investors, and that the acquiring firm's investors then redesign their managerial incentives. Thus the per-unit surplus $\rho_0-P$ goes to the acquiring firm's investors.)

**Context needed to solve it** *From Tirole §14.2.2 "Industry-Wide Shocks and Distress Sales: The Shleifer–Vishny Model", pp. 500–503 (PDF 513–516); equations checked against the PDF images:*

This subsection, building on Shleifer and Vishny (1992), endogenizes the resale price in the two-firm, continuous-investment version of the model of Section 4.3.1. Let us restate the key ingredients of this model.

*Investment and redeployability.* There are two firms in the industry. The "industry" is here defined as a group of symmetric firms using the same equipment/assets. For simplicity, we assume that the two firms do not compete in the same product market (see below for a discussion of this hypothesis). Each firm is run by an entrepreneur, who has initial cash $A_i$.

Initially, firm $i$ invests $I_i$, and therefore borrows $I_i-A_i$ from some *ex ante* competitive lender ("lender $i$"). Then, there is a costless "learning period." At the end of the learning period, each firm learns whether it is "productive" (which has probability $x$), or "unproductive," i.e., "in distress" (which has probability $1-x$). Being unproductive means that the firm will always be unsuccessful regardless of whether the entrepreneur behaves. For example, there may be no demand for the firm's output. Its assets are then useless if left in place. A productive firm is described as in the variable-investment model of Section 3.4 (which, incidentally, corresponds to the case $x=1$).

If both firms are productive, each manages its initial investment. Firm $i$'s profit is either 0 or $RI_i$. Borrower $i$'s private benefit is 0 (if she behaves) or $BI_i$ (if she does not). The associated probabilities of success are $p_H$ and $p_L$, respectively.

If firm $j$ is in distress, it sells its assets, which now have no internal use. We assume that potential buyers outside the industry do not have the knowledge to operate these assets. Only firm $i$ (if it itself is not in distress) can buy it. There has been no initial contract that would specify the transfer price in the case of distress. Rather, this transfer price is determined through bargaining after distress occurs. We will later determine the per-unit transfer price $P$. The entrepreneur in firm $i$ then manages $(I_1+I_2)$ units of assets, and obtains private benefit 0 (if she behaves) or $B(I_1+I_2)$ (if she does not). Similarly, the income is either 0 or $R(I_1+I_2)$. Probabilities of success are $p_H$ if the entrepreneur behaves and $p_L$ if she does not. Firm $i$ has just grown bigger through the acquisition.

As usual, we assume that in the absence of adverse shock ($x=1$), projects are viable only if the borrower behaves,

$$\rho_1 \equiv p_H R > 1 > p_L R + B, \qquad (14.1)$$

and we make a further assumption guaranteeing that loans are finite (even for $x$ close to 1):

$$p_H R < 1 + \frac{p_H B}{\Delta p} \quad \text{or} \quad \rho_0 \equiv p_H \left( R - \frac{B}{\Delta p} \right) < 1. \qquad (14.2)$$

*Loan agreements.* Lender $i$ and entrepreneur $i$ secretly sign at the start a loan agreement specifying the amount of the loan $I_i - A_i$ and the stake $R_{bi}$ of entrepreneur $i$ in the case of success (in the absence of purchase of firm $j$'s assets). [...]

*Summary of timing* (Figure 14.1, transcribed): for each firm $i = 1, 2$: Loan agreement $i \to I_i \to$ learning period $\to$ no distress (prob. $x$) or distress (prob. $1 - x$), with transfer of assets from a distressed firm to the other firm $\to$ $\text{MH}_i$ (moral hazard in firm $i$) $\to$ Outcome $i$.

*Correlation of shocks.* The shocks affecting the demands for the two products may be correlated. We allow for an arbitrary level of correlation. The conditional probabilities (given firm $i$'s state) that firm $j$ is productive or in distress are stated in Table 14.1. For consistency, the parameters must be such that the probability that firm $j$ is productive is $x$:

$$x\mu + (1-x)(1-\nu) = x \iff x(1-\mu) = (1-x)(1-\nu). \qquad (14.3)$$

**Table 14.1**

| when firm $i$ is | firm $j$ productive | firm $j$ in distress |
|---|---|---|
| productive (prob. $x$) | $\mu$ | $1-\mu$ |
| in distress (prob. $1-x$) | $1-\nu$ | $\nu$ |

[...] *Transfer price.* If both firms are in distress (which has probability $(1-x)\nu$), the four participants (entrepreneurs, lenders) receive no *ex post* revenue. If neither is in distress (which has probability $x\mu$), no sale occurs and the model is the standard variable-investment one.

So let us consider the more interesting case in which *firm 1, say, is in distress and firm 2 is not.* We then assume that *lender 1 makes a take-it-or-leave-it offer to lender 2* (see the third remark below for more general bargaining powers). Let $P$ denote the per-unit price demanded by lender 1.

Note that lender 2 must adjust entrepreneur 2's incentive scheme to account for the increased investment and therefore for the increased private benefit from not behaving (now equal to $B(I_1 + I_2)$ instead of $BI_2$). Assume that entrepreneur 2's incentive compatibility constraint is binding in the absence of a purchase ($(\Delta p)R_{b2} = BI_2$), which actually turns out to be optimal. Then, lender 2 must raise entrepreneur 2's income in the case of success by $\delta R_{b2}$ such that

$$(\Delta p)(\delta R_{b2}) = BI_1.$$

So, entrepreneur 2's rent increases by $[p_H B/\Delta p]I_1$ and the transfer price is

$$PI_1 = p_H \left[ R - \frac{B}{\Delta p} \right] I_1 = \rho_0 I_1.$$

The transfer price is simply the pledgeable income:

$$P = \rho_0.$$

[...] From (14.2), we see that $P < 1$. [...] Because firm 1 has the bargaining power, lender 2 actually gains nothing from firm 1's distress, while entrepreneur 2 pockets the extra agency rent $p_H BI_1/\Delta p = (\rho_1 - \rho_0)I_1$.

*Entrepreneurs' expected utility.* Suppose entrepreneur $i$ maximizes her net utility. As usual, the lenders' zero-profit condition implies that the borrower receives the full surplus associated with investment:

$$U_{bi} = [xp_H RI_i - I_i] + [(1-x)(1-\nu)PI_i] + \left[x(1-\mu)\frac{p_H B}{\Delta p} I_j\right]. \qquad (14.4)$$

The first term in brackets in the expression of $U_{bi}$ corresponds to the case in which distress sales are impossible. The second term comes from lender $i$'s revenue from the sales of assets if only firm $i$ is in distress. The third term represents the expected windfall gain from firm $j$'s distress.

We can rewrite (14.4) as

$$U_{bi} = [x\rho_i + (1-x)(1-\nu)\rho_0 - 1]I_i + [x(1-\mu)(\rho_1 - \rho_0)]I_j. \qquad (14.5)$$

*(The book prints $x\rho_i$; it is $x\rho_1$, as the definition of $\alpha$ just below confirms.)*

So let

$$\alpha \equiv x\rho_1 + (1-x)(1-\nu)\rho_0 - 1$$

and

$$\kappa \equiv x(1-\mu)(\rho_1 - \rho_0) > 0.$$

Using (14.4), we can rewrite borrower $i$'s net utility $U_{bi}$ as

$$U_{bi} = \alpha I_i + \kappa I_j. \qquad (14.6)$$

*Borrowing capacity.* Because lender $i$ expects no surplus from the purchase of firm $j$'s assets when the latter is in distress, lender $i$'s expected profit is

$$xp_H(RI_i - R_{bi}) + (1-x)(1-\nu)PI_i - (I_i - A_i) = 0,$$

where $R_{bi}$ is borrower $i$'s income when firm $i$ does not purchase firm $j$'s assets and is successful. Incentive compatibility requires that

$$(\Delta p)R_{bi} \geqslant BI_i,$$

and, as usual, this inequality is satisfied with equality in order to maximize pledgeable income and therefore debt capacity. Using these two equations, firm $i$'s maximal investment is

$$I_i = kA_i,$$

where

$$k = \frac{1}{1 - \rho_0[x + (1-x)(1-\nu)]}. \qquad (14.7)$$

[...] We also see that the multiplier decreases with $\nu$. That is, *a firm's borrowing capacity decreases with the degree of correlation between firms.* A higher correlation means that the assets are less redeployable.

As we noted, the correlation of shocks reduces the desirability of investment ($\partial\alpha/\partial\nu < 0$). Let $\nu^*$ denote the level of $\nu$ (if it exists) such that $\alpha = 0$. We consider two cases:

(a) *Low correlation* ($\nu < \nu^*$). Then $\alpha > 0$ and the firms invest up to their borrowing capacity: $I_i = kA_i$. While each firm's investment is independent of the existence of the other firm, each firm derives a positive externality from this existence in the form of asset redeployability.

(b) *High correlation* ($\nu > \nu^*$). Then $\alpha < 0$, and no firm invests, even though coordinated investment could be profitable (if $\alpha + \kappa > 0$). [...]

*Remark (alternative distributions of bargaining power).* The analysis above, and the rest of Section 14.2 unless otherwise stated, assumes that the acquired firm's investors have full bargaining power and thereby can charge a price equal to the acquiring investors' willingness to pay ($P = \rho_0$). More generally, depending on the two parties' relative bargaining powers, the transaction price can fall anywhere in the range between 0 (the acquired firm's opportunity cost) and $\rho_0$ (the acquiring firm's investors' willingness to pay).

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**15. Liquidity management and acquisitions**

**Source:** Tirole, Ch. 14, Exercise 14.3, p. 515 (PDF 528) · Answer in book: yes, pp. 618–619

**Exercise 14.3 (liquidity management and acquisitions).** Consider the model of Section 14.2.5 when the retooling cost is random. Suppose that this retooling cost is drawn from cumulative distribution function $F(\rho)$ on $[0, \infty)$, with density $f(\rho)$ and monotonic hazard rate ($f(\rho)/F(\rho)$ is decreasing). The level of the retooling cost is privately observed by the potential acquirer (the safe firm). The timing is as described in Figure 14.2.

Assume that the safe firm's entrepreneur and investors ex ante secretly agree on an investment level $I$ and a credit line $L$. This credit line can be used if needed for the acquisition by the entrepreneur and completed by the liquidity, $\rho_0 I$, that can be raised through a seasoned offering that dilutes the initial investors. (Fixing a credit line $L$ of this sort is indeed an optimal policy.)

One will assume that the seller always has the bargaining power ($z = 1$ in the notation of Section 14.2.5) and therefore sets price $P$. Lastly, let $\rho^*$ denote the equilibrium threshold for the retooling cost (that is, assets in equilibrium are acquired and retooled if and only if $\rho \leqslant \rho^*$).

(i) Write the entrepreneur's optimal liquidity management (to this end, follow the steps described in Chapter 5). Show that given (anticipated) equilibrium price $P$, the threshold $\rho^*$ satisfies the "indifference between make and buy" equation:

$$P + \rho^* = 1.$$

(ii) Write the objective function of the risky firm when in distress. Compute the equilibrium price $P$. Note that $P < 1$. What happens to $P$ if for some reason the anticipated level $L$ increases?

(iii) Suppose that the cumulative distribution function $F(\rho)$ converges to a spike at $\bar{\rho}$.[35] Show that $P + \bar{\rho} = 1$, and that $F(\rho^*)$ converges to 1.

[35] While still satisfying the monotone hazard rate property.

**Context needed to solve it** *From Tirole §14.2.2, p. 500 (PDF 513) — per-unit NPV and pledgeable income:*

$$\rho_1 \equiv p_H R > 1 > p_L R + B, \qquad (14.1)$$

$$p_H R < 1 + \frac{p_H B}{\Delta p} \quad \text{or} \quad \rho_0 \equiv p_H \left( R - \frac{B}{\Delta p} \right) < 1. \qquad (14.2)$$

*From Tirole §14.2.5.3 "Bilateral Monopoly: Is Liquidity Hoarding Held Up?", pp. 506–508 (PDF 519–521):*

Let us first describe the model in the case of a single risky firm and a single safe firm. The two firms choose investment sizes $J$ and $I$, respectively. To simplify notation without altering the basic insights, the two firms are identical, except for the probability of distress, which is 0 for the safe firm and $1 - x$ for the risky one. Both are run by risk-neutral entrepreneurs with initial cash on hand $A$ each and protected by limited liability. The timing is summarized in Figure 14.2.

*Figure 14.2 (transcribed).* Risky firm: invests $J \to$ learns whether productive (prob. $x$) $\to$ moral hazard: behaves ($p_H$, no private benefit) or misbehaves ($p_L$, private benefit $BJ$) $\to$ outcome: success ($RJ$) or failure (0). With prob. $1-x$: assets resale to the safe firm. Safe firm: invests $I$, hoards liquidity $L \to$ (if acquisition) pays retooling cost $\rho J$ (on top of acquisition cost $PJ$) $\to$ moral hazard: behaves ($p_H$, no private benefit) or misbehaves ($p_L$, private benefit $BI$ or $B(I+J)$) $\to$ outcome: success ($RI$ or $R(I+J)$) or failure (0).

We further assume that when buying the $J$ units of asset from the risky firm when the latter is in distress, the safe firm must pay a known retooling cost $\rho J$ to adapt these assets to its own production process. Letting $P$ denote the unit acquisition price, the total cost of the acquisition is thus $(P+\rho)J$. To limit the number of cases to be considered, we assume that $\rho \leqslant \rho_0$.

[...] As we want to allow for a wide range of bargaining powers, let us assume that in the case of distress of the risky firm:

- with probability $z$, the risky firm makes a take-it-or-leave-it resale offer to the safe firm;
- with probability $1-z$, the safe firm makes a take-it-or-leave-it purchase offer.

Thus $z$ is a measure of the selling firm's bargaining power.

*Bargaining and the choice of financial muscle.* The case in which the safe firm makes the offer is a no-brainer: it offers 0 (or just above), the opportunity cost of the risky firm's assets when in distress. By contrast, the situation in which the risky firm makes the offer requires more thinking. Let us assume that the amount of liquidity $L$ hoarded by the safe firm (say, the credit line that the firm secures from its bank[24]) is *not* observed by the risky firm. We look for conditions under which the following is an equilibrium:

- the risky firm demands per-unit price $\rho_0$ for its assets;
- the safe firm, anticipating this and knowing that it will be able to raise $\rho_0 J$ on the asset market, hoards liquidity $L = \rho J$.

Making offer $\rho_0$ is clearly optimal for the seller since $\rho_0 J$ is the pledgeable income on the acquired assets and thus the upper bound on what new investors are willing to contribute in a seasoned offering.

Let us next investigate whether it is indeed in the safe firm's interest to hoard liquidity. Let $y=1$ if it hoards the necessary amount ($\rho J$) and $y=0$ if it does not. Noting that liquidity is not needed to acquire the assets when the acquirer makes the offer (since $\rho \leqslant \rho_0$), the safe firm's NPV is then

$$U_b^s = (\rho_1 - 1)I + (1-x)[z[\rho_1 - (\rho_0+\rho)]y + (1-z)(\rho_1-\rho)]J.$$

The second term on the right-hand side is the expected gain from acquisitions, and uses the fact that the total acquisition cost is $(\rho_0+\rho)J$ when the seller makes the offer and only $\rho J$ when the acquirer makes the offer.

The safe firm's investors must break even, and so

$$\rho_0 I + (1-x)[z[\rho_0 - (\rho_0+\rho)]y + (1-z)(\rho_0-\rho)]J = I - A.$$

The investors' breakeven condition yields the investment level $I$:

$$I = \frac{A + [(1-z)(\rho_0-\rho) - z\rho y](1-x)J}{1-\rho_0}.$$

Note that hoarding liquidity ($y=1$) reduces the investment scale. Substituting into the NPV equation,

$$U_b^s = (\rho_1-1)\frac{A + [(1-z)(\rho_0-\rho) - z\rho y](1-x)J}{1-\rho_0} + (1-x)[z[\rho_1-(\rho_0+\rho)]y + (1-z)(\rho_1-\rho)]J.$$

We conclude that it is an equilibrium for the safe firm to hoard liquidity to purchase assets if and only if $U_b^s$ is increasing in $y$, or

$$\rho_1 - (\rho_0+\rho) \geqslant \frac{(\rho_1-1)\rho}{1-\rho_0}$$

or

$$1 \geqslant \rho_0 + \rho.$$

We thus obtain the simple result that the potential acquirer builds the necessary financial muscle if and only if the total per unit cost of acquisition when the acquired firm has bargaining power is lower than the safe firm's own cost of investment. [...]

In this simple model the equilibrium is in general not unique: there are lots of other self-fulfilling equilibria in which the firm hoards $L^* \neq \rho J$ and the seller demands $P^*$ such that $P^* + \rho = L^* + \rho_0$. The seller does not want to demand more than $P > P^*$ because being too greedy prevents the potential acquirer from buying the assets. Conversely, the potential acquirer is willing to hoard $L = L^*$, as long as $P^* + \rho \leqslant 1$ (note that $\rho_0$ is in the interior of the range of equilibrium prices).

[24] We adopt the convention that the safe firm's entrepreneur can raise $\rho_0$ in a seasoned security offering, to which $L$ is added to form total available cash. Alternatively, and as we discussed in Chapter 5, one could ban seasoned offerings and provide the firm with a bigger credit line.

*From Tirole §5.3.1 "The Two-Shock Case", pp. 207–209 (PDF 220–222) — "the steps described in Chapter 5":*

[...] (ii) *Pursue the project even in the case of distress.* The decision to withstand the liquidity shock at date 1 has a cost and benefit. The cost is that the average cost of bringing 1 unit of investment intact to date 2 is $(1 + \lambda\rho)$ (the date-0 cost, 1, plus the expected date-1 reinvestment cost, $\lambda\rho$). The benefit is that the project is never abandoned. The borrowing capacity is given by

$$(1 + \lambda\rho)I - A = \rho_0 I \quad \text{or} \quad I = \frac{A}{(1 + \lambda\rho) - \rho_0}.$$

Similarly, the entrepreneur's utility (the NPV) is

$$U_b^1 = [\rho_1 - (1 + \lambda\rho)]I \quad \text{or} \quad U_b^1 = \frac{\rho_1 - (1 + \lambda\rho)}{(1 + \lambda\rho) - \rho_0} A. \quad [...]$$

[...] If the optimal policy is to pursue the project even in the case of distress, the entrepreneur must be able to avail herself of the amount $\rho I$ if a shock occurs. If $\rho > \rho_0$ [...], then liquidity necessarily must be planned in advance. Waiting exposes the firm to credit rationing at date 1. [...] For example, the firm may contract a credit line to the level of $\rho I$ with a bank; alternatively, it can contract for a credit line corresponding only to the shortfall $(\rho - \rho_0)I$ and also acquire the right to dilute initial investors (so as to obtain $\rho_0 I$).